Midterm — scope and preparation

BioE 147 / BioE 247 · Midterm Thursday 15 October, in class, Dwinelle 219

100 points, the full period. **One page of notes, both sides, prepared by you.** No calculator — every numeric answer is one significant figure, and a bracketing argument is a complete answer.

You write on the exam paper itself, in the box under each item; the figures are printed with the items that use them. Scratch paper is handed out — put your name, SID and the item label on any sheet you want read.

Scope

Sessions 1 through 11 at working depth. You should be able to use this material, not just recognize it.

Session 12 is in scope, lightly. One short item asks you to compare the cell-to-cell variability of two unregulated proteins at different abundances. It needs the scaling, not a derivation.

Session 13 by recognition only. One item marks two noise margins on a supplied transfer curve and asks what quantity equals 1 at the thresholds. It uses **the convention session 13 taught** — the gain as the log slope, thresholds where its magnitude is 1, margins as ratios — and the figure is drawn on log axes so each margin is a distance you can point at. Nothing from that session requires a calculation. You have had one week and no graded practice there, and an exam that tested it at depth would be measuring how fast you read.

Within sessions 1–11, the four things this exam asks you to do:

- **Reason about timescales and magnitudes.** Which of several processes governs a circuit's behavior and which can be neglected; whether a quantity someone hands you is dimensionally possible at all; when an approximation that eliminates a fast variable is legitimate.
- **Turn a mechanism into equations, and say what you would do with them.** Rate laws, stoichiometry, and the assembled system — plus what calculation would answer a question about it, described rather than carried out.
- **Read a dynamical picture.** Nullclines, fixed points and their stability, what the flow tells you, and what happens to the picture when a parameter is changed.
- **Argue about circuit function quantitatively.** Why a circuit is fast or slow, filters or pulses, oscillates or settles — and which parameter you would change to move it from one behavior to the other.

What is not on it

No numeric signal matching between measured gates — that is PS6's job, and its content *is* the computation, which does not fit in an exam period. No delay-differential equations. No derivation reproduced from memory: where a criterion is needed you may bring it on your page, and the credit is in using it.

How it is graded, which should change how you answer

The paper is three parts: **eight short items** worth 36 points, three blocks worth 16 each built on figures, and one longer setup item worth 16 points that asks you to build a system of equations from a reaction list.

Almost every scored part is worth 4 points, and scores 4 / 2 / 0 — 4 for the answer with its reason, **2 for the right answer with no reason or a reason that does not bear on it**, 0 for wrong. That is the half-credit promise below, made concrete: a correct verdict is never worth nothing, and a reason is never worth nothing either.

Three items ask you to work on a figure rather than only in a box, and one of the paper's two biggest items is among them — so leave time for them. Two give you blank axes with time marked and ask you to **sketch output traces** and then mark something on them: shape and relative timing, not numbers. The third asks you to mark two distances on a printed curve. Tuesday's session has you draw a time course yourself, on a different circuit — the drawing *motion* is what we practise, not the exam's particular traces.

On the short items and the blocks, the reason is where the points are. A correct answer with no reason earns half. "Diffusion is faster" is half an answer; "the transcript takes tens of seconds and the protein crosses the cell in tens of milliseconds" is the whole one. A few items are marked *no justification required* — those ask only for a mark on a figure, a definition, or a calculation.

That has a consequence for your note page. **A page that is only a list of formulas will not do much for you.** Every item that could be answered by copying a formula has been written so that it cannot be. Notes that record *when* a result applies, what its terms mean, and what assumption it rests on are worth considerably more than notes recording what the result is.

Preparing

The highest-value preparation is the problem sets you have already done, reworked without looking at your solutions. PS1 through PS5 carry the computational fluency and most of this exam builds on it rather than re-testing it — the exception is the 16-point setup item, where the mechanics are examined directly.

After that: for each of sessions 1–11, write down the one question that session answers, and the one assumption its answer depends on. That second thing is what most of the short items are about.

Bring your page.

Scope questions: the discussion hour Tuesday 13 October, immediately after class in Dwinelle 88 — the last one before the exam.